

Technology Stack

Date	25 September 2026
Team ID	SWTID-2026-6721
Project Name	AI FAQ Assistant API
Maximum Marks	2 Marks

Define Your Technology Stack:

The AI FAQ Assistant API uses a backend-focused technology stack consisting of Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, and Google Gemini AI. The technologies were selected to support secure authentication, FAQ management, database operations, and AI-powered answer generation.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose

1	Frontend / Client-Side	<i>Postman / Thunder Client</i>	<i>The project is a backend REST API without a separate frontend interface. Postman or Thunder Client is used to send requests and test API responses.</i>
2	Backend / Server-Side	<i>Node.js + Express.js</i>	<i>Node.js provides the runtime environment, while Express.js handles API routes, HTTP requests, middleware, and server-side processing.</i>
3	Database / Data Storage	<i>MongoDB + Mongoose</i>	<i>MongoDB stores user and FAQ data. Mongoose provides schemas, validation, and convenient interaction between the Node.js application and MongoDB.</i>
4	Cloud / Hosting / Deployment	<i>Localhost / Node.js Server</i>	<i>The application is currently developed and tested locally using the Node.js server. It can be deployed to a cloud platform in the future.</i>

5	Version Control & CI/CD	<i>Git + GitHub</i>	<i>Git tracks changes in the project, while GitHub stores and manages the source code and project documentation.</i>
6	Third-Party APIs / Other Tools	Google Gemini AI (@google/gemini)	<i>Google Gemini AI is integrated to generate answers to user questions and create AI-assisted FAQ question-answer content.</i>
7	Authentication & Security	<i>JWT + bcrypt + dotenv</i>	<i>JWT protects authenticated routes, bcrypt securely hashes user passwords, and dotenv stores sensitive configuration such as database credentials and API keys in environment variables.</i>
8	API Testing	<i>Postman / Thunder Client</i>	<i>Used to test GET, POST, PUT, and DELETE requests, authentication, FAQ operations, search functionality, and AI endpoints.</i>